Love Plots and Standardized Mean Differences

Thesis Note

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1 Purpose of a Love Plot

A Love plot summarizes covariate imbalance between a treated group and a comparison group. For each covariate X_k, it reports a standardized mean difference, abbreviated as SMD.

In the unadjusted case, the SMD compares the raw covariate means between treated and untreated observations. In the adjusted case, the same comparison is computed after applying estimator-specific weights.

Let $D_i \in \{0, 1\}$ denote treatment status. In the Card application, for example, $D_i = 1$ may indicate that individual i attended some college, while $D_i = 0$ indicates no college attendance.

The key interpretation is:

> The unadjusted Love plot describes the original sample. The adjusted Love plot describes the estimator's weighted sample.

2 Continuous Covariates

Consider a continuous covariate such as experience, denoted by $X_i = exper_i$.

The unadjusted mean among treated observations is

$$\bar{X}_1 = \frac{1}{N_1} \sum_{i:D_i=1} X_i,$$

and the unadjusted mean among untreated observations is

$$\bar{X}_0 = \frac{1}{N_0} \sum_{i:D_i=0} X_i.$$

The unadjusted standardized mean difference is

$$SMD^{unadj} = \frac{\bar{X}_1 - \bar{X}_0}{s_X},$$

where s_X is a standard deviation used to put the difference into standardized units. A common choice is the standard deviation of X_i in the full sample.

Important: for a continuous variable such as experience, the Love plot does not compare only people with exactly five years of experience. It compares the average experience in the treated group with the average experience in the untreated group.

2.1 Numerical Example: Experience

Suppose the treated observations have experience values

$$4, \ 6, \ 8,$$

and the untreated observations have experience values

$$2,\ 3,\ 5.$$

Then the treated mean is

$$\bar{X}_1 = \frac{4+6+8}{3} = 6,$$

and the untreated mean is

$$\bar{X}_0 = \frac{2+3+5}{3} = 3.33.$$

The raw mean difference is therefore

$$\bar{X}_1 - \bar{X}_0 = 6 - 3.33 = 2.67.$$

If the full-sample standard deviation of experience is $s_X = 2.2$, then

$$SMD^{unadj} = \frac{2.67}{2.2} = 1.21.$$

Thus, before weighting, the treated group has average experience that is approximately 1.21 standard deviations higher than the untreated group.

3 Adjusted SMDs: What the Estimator Weights Do

Many estimators can be represented as weighted outcome estimators:

$$\hat{\tau} = \sum_{i=1}^{N} \omega_i Y_i.$$

The weights ω_i describe how much each observation contributes to the final treatment-effect estimate. In the Love plot, these outcome weights are used to ask whether the observations that effectively drive the estimate are balanced in their covariates.

For balance diagnostics, signed outcome weights are usually converted into positive within-group weights. Denote these positive balancing weights by a_i. Then the weighted mean of covariate X_i among treated observations is

$$\bar{X}_1^w = \frac{\sum_{i:D_i=1} a_i X_i}{\sum_{i:D_i=1} a_i},$$

and the weighted mean among untreated observations is

$$\bar{X}_0^w = \frac{\sum_{i:D_i=0} a_i X_i}{\sum_{i:D_i=0} a_i}.$$

The adjusted standardized mean difference is

$$SMD^{adj} = \frac{\bar{X}_1^w - \bar{X}_0^w}{s_X}.$$

The estimator does not necessarily compare each treated person to one specific untreated person. Instead, it changes the relative importance of observations. Some observations receive more weight, others receive less weight. If the estimator gives more weight to untreated observations with high experience and less weight to treated observations with high experience, then the weighted treated and untreated means of experience can move closer together.

3.1 Numerical Example with Weights

Using the same experience values as before, suppose the estimator produces the following positive balancing weights:

Observation	D_i	$X_i = exper_i$	a_i
A	1	4	2
B	1	6	1
C	1	8	0.5
D	0	2	0.5
E	0	3	1
F	0	5	3

The weighted treated mean is

$$\bar{X}_1^w = \frac{2 \cdot 4 + 1 \cdot 6 + 0.5 \cdot 8}{2 + 1 + 0.5} = \frac{8 + 6 + 4}{3.5} = 5.14.$$

The weighted untreated mean is

$$\bar{X}_0^w = \frac{0.5 \cdot 2 + 1 \cdot 3 + 3 \cdot 5}{0.5 + 1 + 3} = \frac{1 + 3 + 15}{4.5} = 4.22.$$

The weighted mean difference is therefore

$$\bar{X}_1^w - \bar{X}_0^w = 5.14 - 4.22 = 0.92.$$

Using the same standard deviation $s_X = 2.2$, the adjusted SMD is

$$SMD^{adj} = \frac{0.92}{2.2} = 0.42.$$

The imbalance decreased from

$$SMD^{unadj} = 1.21$$

to

$$SMD^{adj} = 0.42.$$

This means that the estimator's weights made the treated and untreated groups more similar in average experience.

4 Binary Covariates

For a binary covariate, such as $black_i \in \{0, 1\}$ or a regional dummy $reg661_i \in \{0, 1\}$, the mean has a simple interpretation: it is the share of observations for which the variable equals one.

For example, if $X_i = black_i$, then

$$\bar{X}_1 = \frac{1}{N_1} \sum_{i:D_i=1} black_i$$

is the share of Black individuals among treated observations, while

$$\bar{X}_0 = \frac{1}{N_0} \sum_{i:D_i=0} black_i$$

is the share of Black individuals among untreated observations.

The unadjusted SMD is again

$$SMD^{unadj} = \frac{\bar{X}_1 - \bar{X}_0}{s_X}.$$

For a binary covariate, if the full-sample share is $\bar{X}$, the full-sample standard deviation is

$$s_X = \sqrt{\bar{X}(1-\bar{X})}.$$

4.1 Numerical Example: Binary Covariate

Suppose the binary covariate is $black_i$, and the data are:

Observation	D_i	$black_i$
A	1	1
B	1	0
C	1	0
D	0	1
E	0	1
F	0	0

The treated share is

$$\bar{X}_1 = \frac{1+0+0}{3} = 0.333.$$

The untreated share is

$$\bar{X}_0 = \frac{1+1+0}{3} = 0.667.$$

The raw difference is

$$\bar{X}_1 - \bar{X}_0 = 0.333 - 0.667 = -0.334.$$

If the full-sample share is $\bar{X} = 0.5$, then

$$s_X = \sqrt{0.5(1-0.5)} = 0.5.$$

Therefore,

$$SMD^{unadj} = \frac{-0.334}{0.5} = -0.668.$$

In absolute value, this is

$$|SMD^{unadj}| = 0.668.$$

Thus, the share of Black individuals differs substantially between treated and untreated observations.

5 Adjusted SMD for a Binary Covariate

Now suppose the estimator produces weights such that the weighted share of Black individuals among treated observations is

$$\bar{X}_1^w = 0.40,$$

while the weighted share among untreated observations is

$$\bar{X}_0^w = 0.45.$$

Then the adjusted SMD is

$$SMD^{adj} = \frac{0.40 - 0.45}{0.5} = -0.10.$$

In absolute value,

$$|SMD^{adj}| = 0.10.$$

So after weighting, the treated and untreated groups are much more similar in the share of Black individuals.

6 Binary Region Dummies

The same logic applies to regional indicators. Suppose $reg661_i = 1$ means that individual i comes from a particular region, and $reg661_i = 0$ otherwise.

Then the mean of $reg661_i$ among treated observations is simply the treated share coming from that region:

$$\bar{X}_1 = \Pr(reg661 = 1 \mid D = 1).$$

The mean among untreated observations is

$$\bar{X}_0 = \Pr(reg661 = 1 \mid D = 0).$$

Therefore, the SMD for a region dummy compares regional shares between treated and untreated observations.

6.1 Numerical Example: Region Dummy

Suppose that before weighting,

$$\Pr(reg661 = 1 \mid D = 1) = 0.40,$$

and

$$\Pr(reg661 = 1 \mid D = 0) = 0.05.$$

The raw difference is

$$0.40 - 0.05 = 0.35.$$

If the full-sample share from this region is 0.20, then

$$s_X = \sqrt{0.20(1 - 0.20)} = \sqrt{0.16} = 0.40.$$

Thus,

$$SMD^{unadj} = \frac{0.35}{0.40} = 0.875.$$

This is a large imbalance. It means that treated and untreated observations are very differently distributed across this region.

After weighting, suppose the estimator changes the effective regional shares to

$$\overset{w}{\Pr}(reg661 = 1 \mid D = 1) = 0.25,$$

and

$$\overset{w}{\Pr}(reg661 = 1 \mid D = 0) = 0.22.$$

Then

$$SMD^{adj} = \frac{0.25 - 0.22}{0.40} = 0.075.$$

This would indicate good adjusted balance for this region dummy.

7 Why Regional Dummies Can Remain Imbalanced

In the Card application, the instrument is geographic: college proximity. Therefore, regional variables are difficult to balance because region is strongly related to whether a person lives near a college.

For a regional dummy, adjusted balance requires that the weighted treated and untreated groups contain similar shares of observations from that region. If many treated observations come from a region but only very few comparable untreated observations are available from the same region, the estimator has limited support for constructing a balanced comparison.

In that case, one of two things can happen. First, the adjusted SMD can remain large because the estimator cannot find enough comparison observations in that region. Second, the estimator may assign very large weights to the few available comparison observations, which can reduce imbalance but also create weight concentration and low effective sample size.

Hence, a Love plot should be read together with weight diagnostics such as maximum absolute weights, negative weight shares, and effective sample size. A small adjusted SMD indicates good covariate balance, but if this balance is achieved by putting nearly all weight on a handful of observations, the resulting estimate may still be fragile.

8 Final Interpretation

The unadjusted Love plot asks:

> How different are treated and untreated observations before weighting?

The adjusted Love plot asks:

> How different are the treated and untreated outcome contributions after applying the estimator's weights?

For continuous covariates, the SMD compares weighted or unweighted averages. For binary covariates, the SMD compares weighted or unweighted shares.

Therefore, the Love plot is not a direct matching plot. It does not say that every treated observation is matched to one untreated observation with exactly the same covariates. Instead, it shows whether the estimator's final weighting scheme makes the treated and untreated sides of the comparison similar in observed covariates.